Alex Rivera

Machine Learning Engineer

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Summary

Machine Learning Engineer with 6 years building forecasting, risk and retrieval systems that run in production. I work across the whole path from data pipelines to model serving and monitoring, mostly in Python on AWS. Recent work includes LLM retrieval tools for operations teams and models that cut costs in freight and healthcare.

Skills

Languages & Data: Python, SQL, Scala, Bash, PostgreSQL
Machine Learning: PyTorch, scikit-learn, XGBoost, pandas, Feature Engineering, A/B Testing, Time Series
MLOps & Cloud: AWS, SageMaker, Docker, Kubernetes, MLflow, Airflow, CI/CD, Terraform
GenAI: LLMs, RAG, LangChain, Embeddings, pgvector, LLM Evaluation

Experience

Lumen Freight, USA Mar 2023 – Present
Senior Machine Learning Engineer

- Built a demand forecasting service in PyTorch and Airflow on AWS that predicts load volume for 1,200 routes, cutting empty truck miles by 14% across the regional network.
- Shipped a retrieval tool with LangChain and pgvector that answers dispatcher questions from 40,000 shipping documents, with a 600 question evaluation set run on every release.
- Moved model training to SageMaker with MLflow tracking and Docker images, taking retraining from two days to five hours so the team could release models every week.
- Set up drift monitoring and alerts for 9 production models on Kubernetes, catching two broken data pipelines before they reached the pricing team or customer quotes.

Brightline Health, USA Jun 2020 – Feb 2023
Machine Learning Engineer

- Trained XGBoost models that flag patients at risk of missing follow-up visits, raising scheduled follow-ups by 11% across 23 clinics during a six month pilot.
- Rebuilt the feature pipeline in Python and SQL with tested, versioned transformations, cutting the nightly run from 3 hours to 35 minutes and ending most manual reruns.
- Ran A/B tests on appointment reminder timing with the product team and wrote the analysis templates analysts still use to size and read their experiments.
- Worked with clinicians to define alerts they would trust, and cut false alarms by 30% by adding visit history features and a calibrated decision threshold.

Northwind Analytics, USA Jul 2018 – May 2020
Data Scientist

- Built churn models in scikit-learn for retail clients and explained each model's drivers in plain language so account managers could act on them in quarterly reviews.
- Wrote time series forecasts in pandas for store staffing at 150 locations, reducing overtime hours by 8% in the first year after the forecasts went live.
- Automated weekly client reports with Python and SQL, saving the analytics team about 10 hours a week and removing the copy and paste errors clients used to flag.
- Joined point of sale data from 12 retail clients into one warehouse schema in SQL, which became the base for every model the team built after that.

Education

M.S. Computer Science, Cascade State University Sep 2016 – Jun 2018